

LOCALIQ Architecture & System Definition:

LOCALIQ is an air-gapped, privacy-first on-device knowledge assistant designed to index personal and en

Core Capabilities:

1. Zero-cloud document indexing
2. Privacy-enforced air-gapped knowledge vault
3. Local ONNX neural inference
4. Grounded RAG answer generation

Embedding Model and Vector Dimensionality:

LOCALIQ uses the sentence-transformers/all-MiniLM-L6-v2 embedding model for dense neural representation.

The embedding model produces 384-dimensional dense vectors with unit L2 normalization.

All 384-dimensional vectors are stored locally in the browser IndexedDB vector database scoped by user ID.

This architecture ensures maximum precision and mathematical consistency for cosine similarity comparisons.

Local Semantic Search & Retrieval Pipeline:

LOCALIQ performs semantic search by executing on-device mathematical inner product cosine similarity of

When a user submits a query, LOCALIQ embeds the query with the local all-MiniLM-L6-v2 model, calculates

Every retrieved chunk maintains strict provenance including file name, page number, and chunk index for v